CLIMATE RESILIENCE & ENVIRONMENTAL SUSTAINABILITY

TerraLink

Smarter Wetland Monitoring & Project Management

From a satellite signal to a recorded field response, all in one place.

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terralink-probah.vercel.app



The Problem

Wetland protection requires more than identifying environmental changes.

Ponds, beels and wetlands store rainwater, reduce urban flooding and support fisheries and livelihoods. Around Dhaka, many canals and water bodies have disappeared or become blocked as the city grows.

Environmental project managers and field coordinators face three key challenges:

Difficult monitoring: Comparing satellite images takes time and skill. Clouds or seasonal drying can make a pond look smaller than it is.

Scattered records: Findings, photos, tasks and costs sit in separate files, phones and chat groups.

Limited follow-up: It is unclear who should check a site, by when, and at what cost, so reports take long to prepare.



Urban growth puts steady pressure on ponds and wetlands in and around Dhaka.

The gap A simple workflow that connects environmental observations to the checks, work and costs that follow.

Our Solution

One platform to monitor, review and manage wetland protection projects.

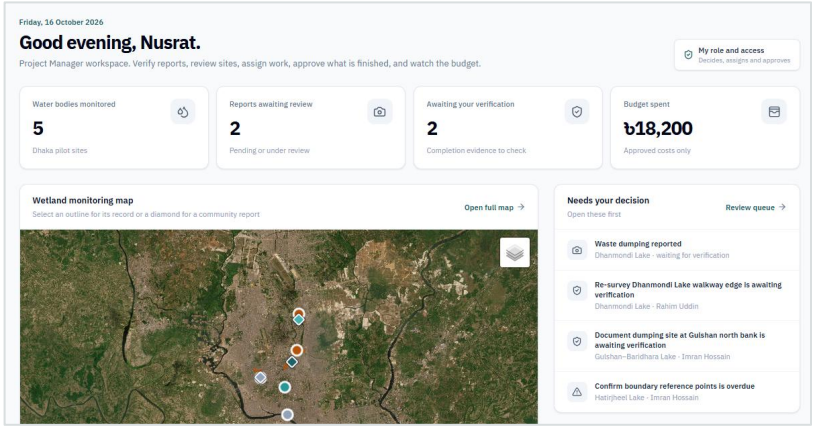
TerraLink keeps one shared record for each site, linking satellite evidence, decisions, tasks and costs.

- Compare satellite images to spot possible changes in visible water area
- Review satellite findings and resident photo reports before deciding
- Assign tasks, deadlines and responsibilities to field coordinators
- Track planned and actual project costs
- Share reports as a PDF by email, with deadlines on the calendar

Built to be trusted

Uses free Sentinel-2 imagery and measures the water area in the browser (NDWI). When clouds hide a site, it shows "Insufficient data" instead of guessing, and a drop in water is treated as a reason to check, not as proof.

Our approach Turn a possible environmental change into a documented, trackable action.



Live prototype: the manager dashboard

Who uses it

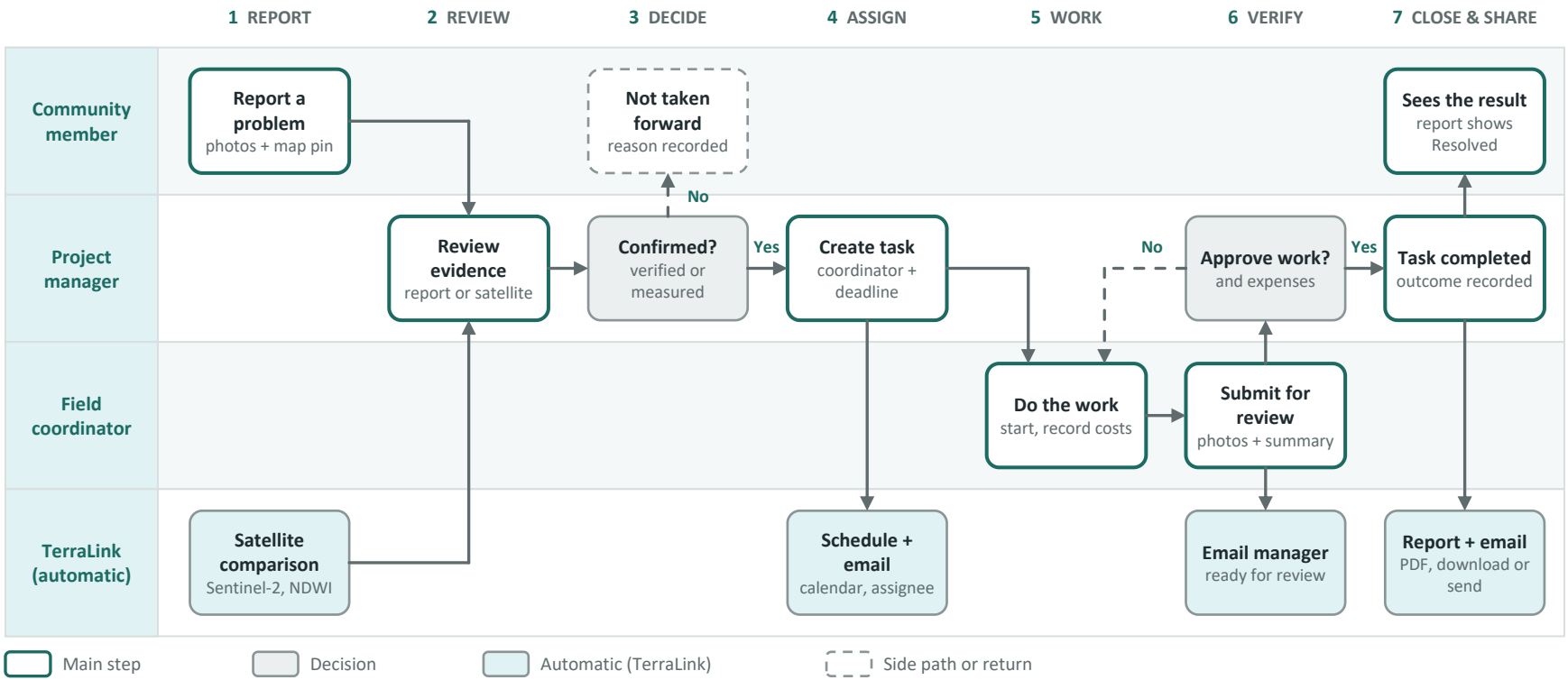
Project Manager reviews findings, approves work, controls budget

Field Coordinator updates assigned tasks and records costs

User explores water bodies and reports problems with photos

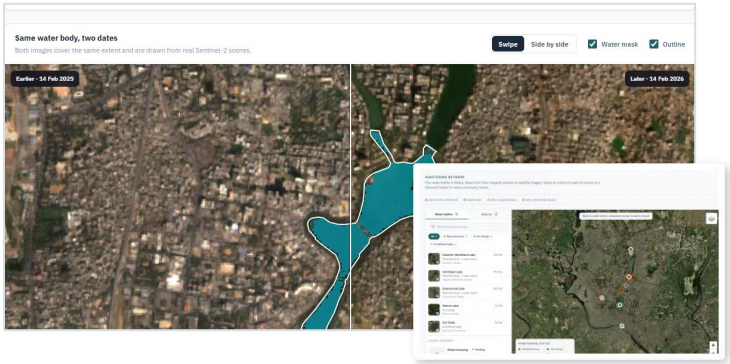
How It Works

From a resident's report or a satellite signal to a closed task and a shared report.



Key Features: Map, Review & Tasks

See the change on real imagery, then turn it into accountable work.



FEATURE 1

Map & Change Analysis

See every site on one map and measure the change between two real images.

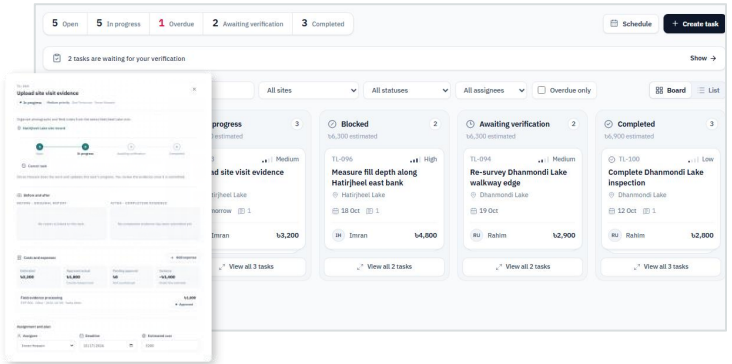
- Real Sentinel-2 scenes, a water mask and the uncertainty stated
- Insufficient data when clouds block the view
- A drop in water is a signal to check, not proof

FEATURE 2

Review & Tasks

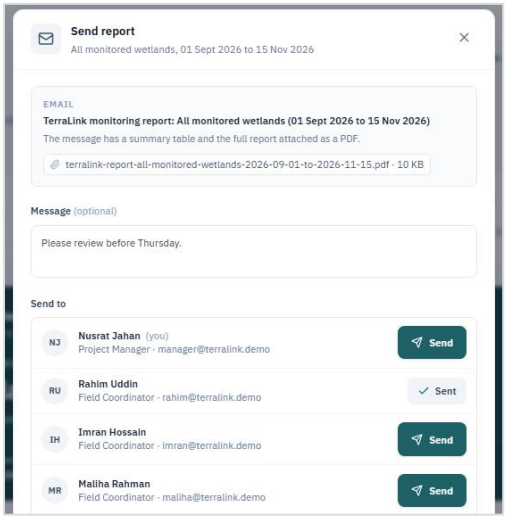
Turn a verified finding into accountable field work.

- Verify resident photo reports and satellite findings
- Assign a field coordinator, deadline and estimated cost
- The coordinator works and submits photos; the manager approves



Key Features: Reports, Calendar & Costs

Reports and events reach the right people without leaving the page where the work happens.



One Send button per person. A tick shows who already has the report.

FEATURE 3

Reports

Download a PDF, or Send: each person gets a subject, a summary table and the PDF attached.

- Images, decisions and costs
- Sources and limits included
- PDF download or email

FEATURE 4

Schedule & Notifications

Tasks and events go on the schedule and Google Calendar, with email invitations and alerts.

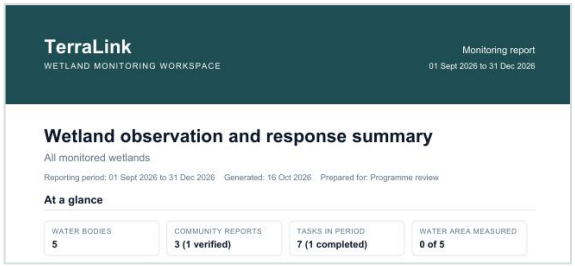
- Events on Google Calendar
- Email invitations, alerts
- Visits next to deadlines

FEATURE 5

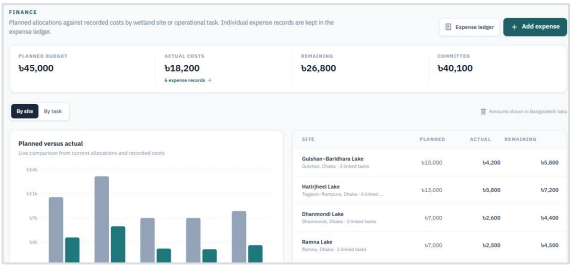
Team & Costs

Planned against actual cost per site; an expense counts once approved, with a budget warning.

- Planned vs actual per site
- Approved expenses only
- Over-budget email warning



The PDF the person downloads



Budgets: planned against actual cost

Key Features: Bangla AI Assistant

Ask in Bangla and get answers from the organisation’s own data, so no one has to dig through the database.

FEATURE 6

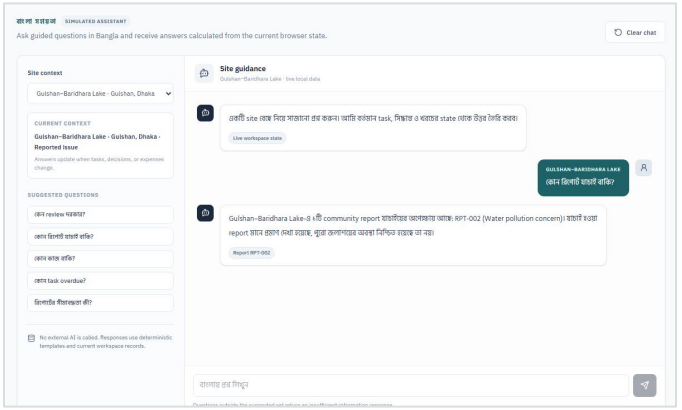


Today: guided answers

- Answers a set list of questions from live project records
- Shows the source record for each answer
- Says clearly when data is missing

Next step: train it on our data

- Train it on the organisation’s own data: past reports, site records, costs and field notes
- Anyone can ask in Bangla, without asking colleagues or searching the database
- Answers stay within what each role may see



Live prototype: a guided assistant that answers from the current records and names its source.

The goal: less time spent searching for information, and fewer people needed to find it.

Expected Impact & Vision

Built for small NGO teams that protect wetlands. Local communities are the ones who benefit.

EXPECTED IMPACT

How will TerraLink help?

One shared record connects residents' reports, satellite evidence, field work and expenses.

- Managers see findings and pending work in one place.
- Report and upload problems with a photo, and follow the result.
- Responsibilities, deadlines and expenses are easy to track.
- Reports go out by email as a PDF.

Actual impact will be evaluated through user feedback and project results.



A focused first step

Five selected Dhaka ponds and wetlands, built on existing satellite data and free tools, with sample records where real data is not available.

Our vision

A practical tool that turns satellite observations into organised, accountable wetland protection work.





Thank You

Questions are welcome.

TerraLink helps NGOs protect wetlands with clear evidence, clear responsibility and clear records.

Live demo

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